

instruments - xsm and class

Documentation: CLASS and XSM Instruments & Data Products

A key concept for both instruments is the **FITS (Flexible Image Transport System)** file format. This is the standard in astronomy for storing scientific data. A FITS file is like a container that can hold multiple components, called **Header Data Units (HDUs)**. Each HDU has two parts:

1. **Header:** A human-readable text section containing metadata (information *about* the data) stored as `KEYWORD = value` pairs.
 2. **Data:** The actual scientific measurements, typically stored as an image or a binary table.
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CLASS (Chandrayaan-2 Large Area Soft X-ray Spectrometer)

What CLASS Measures

CLASS is an X-ray spectrometer designed to look at the **Moon**. Its primary purpose is to perform **X-ray Fluorescence (XRF) spectroscopy** on the lunar surface. It measures the energy of incoming X-ray photons originating from the Moon. By counting how many photons arrive at each specific energy, it builds a spectrum. The peaks in this spectrum correspond to the characteristic "fingerprint" energies of different elements, allowing scientists to determine the elemental composition (Mg, Al, Si, Ca, Ti, Fe) of the lunar regolith.

Data Type and Storage

The fundamental data product from CLASS is a **spectrum**, which is a histogram of X-ray counts versus detector channel (which corresponds to energy). Each spectrum file represents a short observation (typically 4-8 seconds) of a specific area ("footprint") on the lunar surface. All data is stored in the FITS format.

FITS File Structure

You worked with the **Level-1 calibrated data**, which are `.fits` files that can be read as PHA (Pulse Height Analyzer) spectral files.

- **Header:** The header of a CLASS FITS file contains a wealth of crucial metadata. Key keywords include:
 - `TELESCOP` / `INSTRUME` : Identifies the instrument as 'CHANDRAYAAN-2' / 'CLASS'.
 - `EXPOSURE` : The duration of the observation in seconds.
 - `STARTTIME` / `ENDTIME` : The precise start and end time of the observation.
 - `BORE_LAT` / `BORE_LON` : The latitude and longitude of the center of the instrument's footprint.
 - `VO_LAT` , `VO_LON` , etc.: The coordinates of the four corners of the footprint.
 - `SOLARANG` : The solar angle, critical for determining if the surface was illuminated.
 - `GAIN` : The conversion factor from detector channel to energy (in eV/channel).
- **Data:** The data is stored in a `BINTABLE` (Binary Table) extension, typically named `SPECTRUM` . This table contains columns for `CHANNEL` (the detector channel number, from 0 to 2047) and `COUNTS` (the number of X-rays recorded in that channel).

Calibration Files

- **ARF (`class_arf_v1.arf`):** The **Ancillary Response File**. This is the "filter" or sensitivity file. It describes the instrument's effective collecting area (in cm²) as a function of energy. It tells you how efficient the detector is at seeing photons of different energies.
- **RMF (`class_rmf_v1.rmf`):** The **Response Matrix File**. This is the "blur" file. It describes the instrument's energy resolution—how a photon of a single, perfect energy gets spread out over several detector channels.
- **Background File (`background_allevvents.fits`):** This is a pre-made spectrum created by averaging many CLASS observations of the dark, night side of the Moon. It

represents the persistent instrumental noise or "hum" that must be subtracted from any scientific observation.

Data Usage in This Project

- **Calibrated Data (`.fits` / `.pha`):**
 - Headers were read by `perMonthEachDay_extract_class_metadata.py` to create the initial data catalog.
 - The full `COUNTS` and header data were used in the `run_continuum_subtraction` function (inside `run_analysis_batch.py`) to perform the scientific analysis.
 - **ARF File (`class_arf_v1.arf`):**
 - Used by the `load_arf_interpolator` function (in `py_forward_model_and_subtract.py`) to correctly model the strength of the scattered solar continuum on the CLASS detector.
 - **Background File (`background_allevts.fits`):**
 - Used by the `load_background` function (in `py_forward_model_and_subtract.py`) and subtracted from the total observed signal to help isolate the XRF peaks.
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XSM (Solar X-ray Monitor)

What XSM Measures

XSM is also an X-ray spectrometer, but it is designed to look continuously at the **Sun**. Its purpose is to measure the full spectrum of solar X-rays in the 1-15 keV range with high time resolution (a new spectrum every second). This provides the crucial "source" spectrum that illuminates the Moon and causes the fluorescence that CLASS measures.

Data Type and Storage

The primary data product is a **time-series of spectra**. For each second of the day, a full spectrum is generated. This is all stored in a single, large FITS file for each day.

FITS File Structure

You worked with the **Level-1 calibrated data** (`ch2_xsm_..._level1.fits`).

- **Header:** The primary header contains metadata about the entire day's observation, such as `TELESCOP` , `INSTRUME` , `DATE-OBS` , and `DATE-END` .
- **Data:** The data is a single, large `BINTABLE` extension containing approximately 86,400 rows (one for each second in a day). The key columns we used are:
 - `UTCString` : A human-readable timestamp for each row.
 - `dataArray` : The actual spectrum for that second, stored as an array of 2048 numbers representing the counts in each channel.

Calibration Files

- **RSP (`CH2xsmrspwitharea_open20191214v01.rsp`):** The **Response File**. This convenient `.rsp` file contains *both* the RMF ("blur") and ARF ("filter"/sensitivity) information for the XSM instrument in a single file.
- **Background File (`ch2_xsm_20200128_bkg.pha`):** A pre-made spectrum representing the instrumental background for the XSM detector, which needs to be accounted for during the fit.

Data Usage in This Project

- **Calibrated Data (`..._level1.fits`):**
 - Read by `new_extract_xsm_metadata.py` for initial data cataloging.
 - The `DataArray` counts were read by `prepare_xsm_for_xspec.py` to create the time-averaged `.pha` files for analysis.
- **RSP and Background Files:**
 - Both files were used by the `run_xspec_analysis_with_pyxspect` function (inside `run_analysis_batch.py`). They are essential inputs for the **forward-fitting** process in `XSPEC` , allowing it to remove the instrumental effects and determine the true, physical solar spectrum `F_sun(E)` .